

Xtra – Some Mathematical Simplifications

Q) Why would $O(\log_2(N + 1)) = O(\log_2 N)$?

$$\begin{aligned} & \log_2(N + 1) \\ &= \log_2\left(N\left(1 + \frac{1}{N}\right)\right) \end{aligned}$$

Using the logarithm law,

$$\log_2(ab) = \log_2 a + \log_2 b$$

we get,

$$\log_2(N + 1) = \log_2 N + \log_2\left(1 + \frac{1}{N}\right)$$

Note,

$$\text{As, } N \rightarrow \infty$$

$$\lim_{n \rightarrow \infty} \left(\log_2\left(1 + \frac{1}{n}\right) \right)$$

if $\lim_{u \rightarrow b} f(u) = L$, and $\lim_{x \rightarrow a} g(x) = b$ and $f(x)$ is continuous at $x = b$

Then:

$$\lim_{x \rightarrow a} f(g(x)) = L$$

$$g(n) = 1 + \frac{1}{n} \text{ and } f(u) = \log_2(u)$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)$$

$$= \lim_{n \rightarrow \infty} (1) + \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)$$

$$= 1 + \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right) \left[\lim_{x \rightarrow a} (c) = c, \text{ where } c \text{ is a constant} \right]$$

$$= 1 + 0 \left[\lim_{x \rightarrow \infty} \left(\frac{c}{x^a} \right) = \frac{c}{\infty} = 0, \text{ where } c \text{ is a constant, } a > 0 \right]$$

$$= 1$$

$$\lim_{u \rightarrow 1} (\log_2(u)) = \log_2(1) = 0 \quad [\log_a(1) = 0]$$

$$\therefore \lim_{n \rightarrow \infty} \left(\log_2 \left(1 + \frac{1}{n} \right) \right) = 0$$

It means : $\log_2 \left(1 + \frac{1}{n} \right)$ converges to 0 .

$$\log_2(N + 1) = \log_2 N + \underbrace{\log_2 \left(1 + \frac{1}{N}\right)}_{\rightarrow 0}$$

Since the extra term converges to 0, it is bounded (in fact, it is $o(1)$). Therefore,

$$\log_2(N + 1) = \log_2 N + o(1)$$

and consequently,

$$O(\log_2(N + 1)) = O(\log_2 N)$$

We have seen :

$$S_1 = \sum_{i=0}^k i \times 2^i$$

$$S_1 = 0(2^0) + 1(2^1) + 2(2^2) + 3(2^3) + \dots + k-1(2^{k-1}) + k(2^k)$$

$$\frac{2}{1} = 2, \frac{4}{2} = 2, \frac{8}{4} = 2, \frac{16}{8} = 2, \dots, \frac{2^k}{2^{k-1}} = 2^{k-(k-1)} = 2^1 = 2,$$

common ratio: 2

Multiplying both sides by common ratio `2`, we get:

$$2S_1 = 0(2^1) + 1(2^2) + 2(2^3) + \dots + k-1(2^k) + k(2^{k+1})$$

We can rewrite it as :

$$2S_1 = 0(2^0) + 0(2^1) + 1(2^2) + 2(2^3) + \dots + (k-2)2^{k-1} \\ + k-1(2^k) + k(2^{k+1})$$

Now, subtract this new equation ($2S_1$) from the original equation (S_1):

$$S_1 - 2S_1 = (0-0)2^0 + (1-0)2^1 + (2-1)2^2 + (3-2)2^3 \\ + \dots + (k-1-(k-2))2^{k-1} + (k-(k-1))2^k \\ + (0-k)2^{k+1}$$

$$s_1 - 2S_1 = 0 + 2 + 4 + 8 + \dots + 2^{k-1} + 2^k - k2^{k+1}$$

$$\Rightarrow -S_1 = 0 + 2 + 4 + 8 + \dots + 2^{k-1} + 2^k - k2^{k+1}$$

$(0 + 2 + 4 + 8 + \dots + 2^{k-1} + 2^k)$ represents geometric series: $1 \times 2^1 + 1 \times 2^2 + 1 \times 2^3 + \dots + 1 \times 2^{k-1} + 1 \times 2^k$

$$S(n) = ar^1 + ar^2 + \dots + ar^n$$

then this is a geometric series with:

- First term: ar
- Common ratio: r

Multiply by r :

$$rS(n) = ar^2 + ar^3 + \dots + ar^{n+1}$$

Now subtract:

$$S(n) - rS(n) = (ar^1 + ar^2 + \dots + ar^n) - (ar^2 + ar^3 + \dots + ar^{n+1})$$

$$= (ar^1 + ar^2 - ar^2 + ar^3 - ar^3 \dots + ar^n - ar^n + 0 - ar^{n+1})$$

$$= (ar^1 - ar^{n+1})$$

$$S(n) - rS(n) = (ar - ar^{n+1})$$

$$(1 - r)S(n) = (ar - ar^{n+1})$$

$$S(n) = \frac{(ar - ar^{n+1})}{1 - r}$$

$$\text{or, } S(n) = \frac{ar^{n+1} - ar}{r - 1} \left[\text{Multiplying } -1 \text{ in both numerator and denominator, we get} \right]$$

$$\text{or, } S(n) = \frac{ar(r^n - 1)}{r - 1}$$

$$S(n) = a \left(\frac{r(r^n - 1)}{r - 1} \right) = 1 \left(\frac{2(2^k - 1)}{2 - 1} \right) = 2^{k+1} - 2.$$

$$\therefore -S_1 = 2^{k+1} - 2 - k2^{k+1}$$

To cancel out -1 , Multiplying -1 in both the sides:

$$\therefore S_1 = -2^{k+1} + 2 + k2^{k+1}$$

$$= k2^{k+1} - 2^{k+1} + 2$$

$$= 2^{k+1}(k - 1) + 2$$

$$\therefore S_1 = 2^{k+1}(k - 1) + 2$$

Similarly, if there is :

$$S_1 = \sum_{i=0}^k i \times 3^i$$

The common ratio of the geometric part : $\frac{3^{i+1}}{3^i} = 3$.

$$\text{Hence : } S_1 - 3S_1$$

$$\text{When , } S_1 = \sum_{i=0}^k i \times 4^i$$

The common ratio of the geometric part : $\frac{4^{i+1}}{4^i} = 4$.

$$\text{Hence : } S_1 - 4S_1$$

.

.

.

$$\text{When , } S_1 = \sum_{i=0}^k i \times n^i$$

The common ratio of the geometric part : $\frac{n^{i+1}}{n^i} = n$.

$$\text{Hence : } S_1 - nS_1$$

The common ratio of the geometric part : $\frac{n^{i+1}}{n^i} = n$.

$$\text{Hence : } S_1 - nS_1$$

Why so?

$$S_1 = 0 \times n^0 + 1 \times n^1 + 2 \times n^2 \dots + k \times n^k$$

$$nS_1 = n(0 \times n^0 + 1 \times n^1 + 2 \times n^2 \dots + k \times n^k)$$

Each term becomes:

$$0 \cdot n^0 \times n = 0 \cdot n^1$$

$$1 \cdot n^1 \times n = 1 \cdot n^2$$

$$2 \cdot n^2 \times n = 2 \cdot n^3$$

.

.

$$k \cdot n^k \times n = k \cdot n^{k+1}$$

Therefore,

$$nS_1 = \sum_{i=0}^k i \times n^{i+1}$$

$$\text{And, } S_1 = \sum_{i=0}^k i \times n^i$$

$\therefore$ The common ratio of the geometric part : $\frac{n^{i+1}}{n^i} = n$.
